

WORDS WHICH GIVE RISE TO ANOTHER GROUP
OPERATION FOR A GIVEN GROUP

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1. Introduction

For any group G , the binary operation $x * y = yx$ is a group operation, expressed in terms of the word yx , which gives a different (unless G is abelian) but isomorphic group structure on the set G . It is natural to ask what other group structures may be defined on G by an operation of the form

$$x * y = W(x, y)$$

where $W(x, y)$ is a word in x, y (we call such words *group words* for G) and whether it is possible for the new group (denoted by G_W) and G to be non-isomorphic. If so, it might be possible to discover facts about one group by considering the other - especially if one group is abelian and the other is not. Thoughts such as these have prompted this paper.

Most of the work that has been done previously on group words has been restricted to what we may call *non-singular* group words - those for which $G = \{G_W\}_Y$ for some group word Y (evaluated in G_W). Higman and Neumann [8] asked whether $G \cong G_W$ for all non-singular group words W and Hulanicki and Swierczkowski [9] showed that this is true for periodic nilpotent groups of class 2. More recently, Street [11]* not only extended this to "most" nilpotent groups of class 3 and 4 (including, for example, all finite p -groups of class at most 4 for $p \geq 5$) but also gave an infinite family of counter-examples, the smallest of which has order 602.

In this paper we investigate non-abelian groups G for which there is a group word W (necessarily *singular*) for which G_W is abelian. We call such groups *verbally abelian*. Baer [2] pointed out a long time ago that if G is a nilpotent group of class 2 such that G' has finite odd exponent n and

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$W(x, y) = xy[x, y]^{(n-1)/2}$, then G_W is abelian. This fact proved useful in a paper of Bender [3]. We extend this here to nilpotent groups of class 3. In addition we discuss, for a general group G , the relationship between the properties of G and G_W .

2. Nilpotent groups of class at most 3

Similar calculations to those of Street [11] yield the following.

THEOREM 1. *The group words for a nilpotent group G of class at most 3 are those which are equivalent to (that is, are identical to when evaluated in G)*

$$(1) \quad xy[x, y]^r [x, y, x]^{-s} [x, y, y]^s$$

for some integers r, s satisfying the congruence

$$(2) \quad 3s \equiv r^2 + r \pmod{\text{expt}[G', G]}.$$

If W is such a group word for G , G_W is abelian if and only if

$$(3) \quad 2r + 1 \equiv 0 \pmod{\text{expt } G'}.$$

COROLLARY. *If G is a nilpotent group of class at most 3 then G is verbally abelian if and only if $2 \nmid \text{expt } G'$ and $3 \nmid \text{expt}[G', G]$.*

From this corollary we see that if G is a finite nilpotent group of class at most 3 then it is verbally abelian if and only if its Sylow subgroups are regular p -groups. There are, however, regular p -groups of class 4 which are not verbally abelian. For example, if $p \geq 7$ and G is the free product of two cyclic groups of order p then it can be shown that $G/[G', G, G, G]$ is not verbally abelian, yet it is a finite regular p -group of class 4.

If G_W is abelian and the operation is written additively, one asks what words may be used to define a multiplication making G into a ring. A related question was answered in the case of associative rings coming from class 2 groups in [1], but, as the next theorem shows, it is more natural for this question to ask for a Lie ring rather than an associative one.

THEOREM 2. *Suppose G is a nilpotent group of class at most 3 for which an abelian group operation, $+$, is given by (1), (2), (3). The words which produce an operation which is distributive over $+$ are those equivalent to*

$$[x, y]^t [x, y, x]^{rt} [x, y, y]^{rt}$$

for some integer t . Moreover, for any such word, G becomes a Lie ring.

3. Relationship between G and G_W

The following theorem suggests that G_W is at least as "nice" a group as G and leads one to make the following conjecture:

CONJECTURE G_W is contained in the variety generated by G .

THEOREM 3. Suppose G is a group and W is a group word for G . Then

- (i) (normal) subgroups of G are (normal) subgroups of G_W ,
- (ii) $\text{aut}(G) \leq \text{aut}(G_W)$,
- (iii) elements have the same order in G as in G_W ,
- (iv) the descending central series and derived series descend at least as quickly in G_W as in G and the ascending central series and Fitting series ascend at least as quickly in G_W as in G ,
- (v) if G is abelian, nilpotent or soluble so is G_W .

These facts are not surprising and are easily proved (cf. Street [11], Theorem 1). A little less obvious are the following which proceed in the opposite direction.

THEOREM 4. Suppose G is a group and W is a group word for G . Then

- (i) if G is locally nilpotent, characteristic subgroups of G_W are characteristic subgroups of G ,
- (ii) if G is finite, G and G_W have the same Sylow subgroups and the same Fitting series,
- (iii) if G is finite, and G_W is nilpotent or soluble then so is G .

4. Finite verbally abelian groups

Suppose that G is a finite verbally abelian group. Theorem 4 (iii) shows that G must be nilpotent and Theorem 1 (Corollary) gives examples of verbally abelian groups up to class 3. Theorem 3 (iii) shows that the abelian group G_W must be unique up to isomorphism. The following facts are obvious for finite abelian groups and may be extended to finite verbally abelian groups using Theorems 3 and 4.

THEOREM 5. Suppose G is a finite verbally abelian group. Then

- (i) for all integers n , $\{g \mid g^n = 1\}$ and $\{g^n \mid g \in G\}$ are characteristic subgroups of G , and all elements which have n 'th roots have the same number of them,

- (ii) if θ is an automorphism of G such that $H^\theta = H$ for all subgroups H , then θ has the form $x \mapsto x^n$ for some integer n ,
- (iii) G is conformal (that is, has the same number of elements of each order as) some abelian group.

We now return to the original motive for studying group words - knowing something about G_W we might be able to translate it back to the "not-so-nice" group G . We have done this to a small extent in being able to show by Theorem 1 (Corollary) that (i) - (iii) of Theorem 5 hold for finite p -groups of class less than $\min(p, 4)$. However such groups are precisely the finite regular p -groups of class at most 3 and (i) - (iii) of Theorem 5 are already known for all finite regular p -groups (see [7] for (i), [5] for (ii), [4] for (iii)). So purely as a tool, group words have not (as yet) justified their existence. However they raise a number of interesting questions about themselves. For example, the fact that finite verbally abelian p -groups have very similar arithmetic properties to finite regular p -groups suggests that perhaps they are in fact always regular (as we pointed out above, the converse is false).

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